

Sarah Hyunju Song (송현주)

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SUMMARY

Backend and full-stack software engineer on a trading systems team at Nomura Securities, where I joined as an intern and have since spent two years building and operating the systems that keep trading compliant and reliable. I work mainly on pre- and post-trade controls and trade validation, collaborating with global teams across New York, London and Tokyo. I enjoy finding problems that keep recurring and solving them with the right technology.

EXPERIENCE

Nomura Securities International, Inc.

New York, NY

Software Engineer

June 2024 – Present

Java, Spring, FastAPI, LangChain, Angular, Kafka, Apache Pulsar, Kubernetes, Helm, FluxCD

- Designed and implemented new trade validation rules across pre-trade and post-trade control systems handling millions of trades daily, partnering with risk and compliance teams to turn business and regulatory requirements into controls that block non-compliant orders before execution and surface violations after it.
- Migrated the deployment pipeline from Jenkins and Ansible to a GitOps workflow built on Kubernetes, FluxCD and Helm, reducing deployment time and establishing a scalable release environment.
- Built a knowledge-based RAG chatbot and an MCP server to automate production support using LangChain, FastAPI and Angular, cutting internal query turnaround from one hour to under ten minutes, with LangSmith for monitoring.
- Modernized the messaging layer by replacing Kafka with Apache Pulsar, adding geo-replication for regional failover and reducing operational costs by over 66%.
- Developed and operated full-stack web applications for internal trade controls, including trade hard block removal and counterparty validation, using Angular and Spring Boot.

Software Engineering Intern

June 2023 – August 2023

Java, Spring Boot, Elasticsearch

- Built a data integrity validation system for historical trade data with Spring Boot and Elasticsearch, re-querying stored records through an API and detecting discrepancies in an automated validation pipeline.
- Implemented a multithreaded processing engine, improving data throughput by 48%.

University of Arizona, CLU Lab

Tucson, AZ

Research Assistant

May 2023 – May 2024

Python, React, TypeScript, PyTorch, TensorFlow

- Designed an AWS-based LLM classification pipeline that automatically improved metadata quality for 78% of scholarly records.
- Designed and built a search and discovery platform for scholarly records using React and TypeScript.
- Developed named entity recognition models with TensorFlow and PyTorch, improving baseline performance by 12%.

EDUCATION

University of Arizona — M.S., Computer Science

2022 – 2024

University of Arizona — B.S., Computer Science, Magna Cum Laude (3.89 GPA)

2020 – 2022

CERTIFICATIONS & LANGUAGES

AWS Certified Cloud Practitioner

December 2025

English (Native), Korean (Native)